



ClipCap+ +: An efficient image captioning approach via image encoder optimization and LLM fine-tuning

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HIGHLIGHTS

- Propose an image captioning method with high accuracy and generalization ability.
- Improve ClipCap by integrating key-value pair and residual connection modules.
- Use hard and soft prompts to fine tune LLM for generating image descriptions.

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ABSTRACT

ClipCap (CLIP prefix for image captioning), a leading image captioning model, exhibits limitations in recognizing images within specific domains. This study presents ClipCap+ +, an enhanced version of ClipCap that integrates key-value pair and residual connection modules. The key-value pair module implements a few-shot learning strategy by incorporating domain-specific knowledge, thereby improving the model's capability to recognize specialized image categories. The residual connection module optimizes the weight distribution between the pre-trained model and the key-value pair module, enhancing the model's transfer learning performance. During the inference phase, the model processes an input image through a multi-stage pipeline: (1) the visual encoder extracts image features to generate a hard visual prompt, (2) the key-value pair module dynamically constructs a domain-specific soft prompt, and (3) these complementary prompts are jointly fed into the large language model to synthesize the final image description. Extensive experiments on in-domain, near-domain, and cross-domain tasks show ClipCap+ + surpasses state-of-the-art models in accuracy, training efficiency, and generalization.

1. Introduction

The goal of image captioning is to describe the key information (rather than background information or noise) conveyed by an image using accurate and meaningful sentences. The challenge of image captioning lies in the alignment between images and text, which is deeply rooted in the image-text pairs. The ClipCap [1] model generates meaningful captions by training a mapping network that bridges the semantic gap between the CLIP (contrastive language-image pre-trained) [2] model and the GPT-2 (generative pre-trained Transformer 2.0) [3]. Fig. 1 shows the architecture of the ClipCap model, where “Const” is a learnable constant. ClipCap introduces a lightweight mapping network to transform the image embeddings generated by CLIP into prompt sequences recognizable by the language model. The prompt

sequences are then concatenated with the caption embedding sequence generated by the language model and fed into GPT-2 to generate natural language descriptions of the image content.

The sources of data for image captioning technology primarily fall into two categories: utilizing existing image-text datasets or creating image-text pairs from independent sources. The former relies on deep learning techniques [4–8], necessitating substantial annotated data for training. Challenges such as long-tail distributions, annotation noise, and high labeling costs hinder the collection of image-text pairs. Long-tail distributions result in sparse samples for certain categories, making it difficult for models to learn their features and thereby affecting generalization. Noisy annotations introduce erroneous information that misleads model training. High annotation costs restrict the scale and diversity of datasets, impacting model training effectiveness

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and generalization. The latter employs mapping network approaches [9–11], utilizing pre-existing multimodal models and large-scale language models as encoders and decoders. However, these methods often yield lower-quality descriptions and exhibit exposure bias in specific tasks. Thus, synthetic data offers a cost- and time-efficient alternative to manual labeling or web crawling. Nevertheless, it exhibits poorer cross-domain image captioning capabilities and lower accuracy in generating text descriptions within specific fields. Currently, there is no dedicated approach for multimodal pre-trained and large-scale language model integration specifically tailored for image captioning in specific domains.

To address these challenges, this paper proposes a novel image captioning model, ClipCap++, which integrates multimodal pre-trained and large-scale language models to streamline the caption generation process. Building upon the ClipCap model, we introduce key-value pair modules and residual connection modules to enhance the accuracy of image captioning tasks within specific domains. Specifically, we utilize the contrastive language-image pre-trained encoder, CLIP, significantly reducing training resources and time costs. After CLIP pre-trained, we incorporate information from a few-shot dataset stored in the key-value pair module. Residual connections are employed to link pre-trained visual features with optimized features, mitigating overfitting to few-shot data. Quantitative analysis demonstrates that our model surpasses the performance of the state-of-the-art ClipCap model in caption generation capability, with comparable training times and parameter scales. Compared to current deep learning-based image captioning models, our approach achieves orders of magnitude reduction in training time. In summary, our contributions are as follows:

- We propose a rapid and efficient lightweight image captioning generation technique that leverages transfer learning by freezing multimodal pre-trained models and large language models to generate high-quality image-text pairs.
- By innovatively integrating key-value pair module and residual connection module, we significantly enhance the widely used ClipCap model, improving its comprehensive understanding of image scenes and generating more accurate and meaningful text descriptions.
- Extensive experiments on in-domain, near-domain, and cross-domain image captioning tasks demonstrate the significant advantages and excellent generalization ability of our proposed image captioning method across various scenarios.

2. Related works

Images and text are primary means of acquiring visual information. Images are vivid and intuitive, leaving deep impressions, while text is concise and clear, precisely depicting and conveying information. With the continuous advancement of image classification, object detection, and recognition technologies, computers' ability to understand images is

constantly improving. Simultaneously, natural language processing techniques are also refining, making it possible to generate textual descriptions from images.

Current image captioning techniques are mainly divided into four categories: retrieval-based methods, template-based methods, deep learning-based methods, and mapping network-based methods. Retrieval-based methods [12,13] include traditional keyword-based retrieval and recent semantic retrieval methods based on vector space models (such as Word2Vec [14] and BERT [15]). These methods require a fixed title-sentence library and match sentences by extracting image features. However, the precision of similarity calculations limits the generated descriptions, making them relatively constrained. Template-based methods [16] use the Curran and Clark parser to generate sentences based on templates. While this approach can increase sentence diversity, the generated content is often fixed and singular.

Deep learning-based methods include the CoCa (contrastive captioners) [17] model for contrastive learning, the Frozen [18] model for unified vision-language pre-trained, and the GRIT (grid- and region-based image captioning Transformer) [19] model employing dual visual features with an efficient Transformer [20] architecture. These methods enhance the quality and speed of descriptive caption generation by combining complex training frameworks and model architectures. However, they typically require substantial computational resources and extensive pre-trained on large datasets, which may limit their accessibility and practical deployment. Additionally, they introduce challenges such as long-tail distributions, noisy annotations, and increasing annotation costs. The long-tail distribution, with its scarcity of samples in certain categories, makes it difficult for models to learn features of these rare classes and affects generalization. Noisy labels introduce erroneous information that misleads model training, especially in the long-tail categories with limited samples. Meanwhile, high annotation costs restrict the scale and diversity of datasets, further impacting model training outcomes and generalization abilities.

With the advent of large language models, image captioning technology has entered a promising phase, achieving significant progress through the synergy of natural language processing and computer vision. Large language models, such as the GPT series [21], offer robust language generation capabilities through pre-trained on extensive text data, injecting new vitality into image captioning tasks. Mapping network-based methods address these issues by adopting transfer learning concepts. They leverage pre-trained models for feature encoding, keeping model parameters frozen during vision-language pre-trained, thus reducing training costs. Aligning visual features with the textual space is challenging because visual information is high-dimensional and continuous, while text is discrete and sequential. There is a significant difference in dimensionality and semantic expression between the two, and effective mapping is required to preserve semantic integrity. Models such as ClipCap, BLIP-2 (bootstrapping

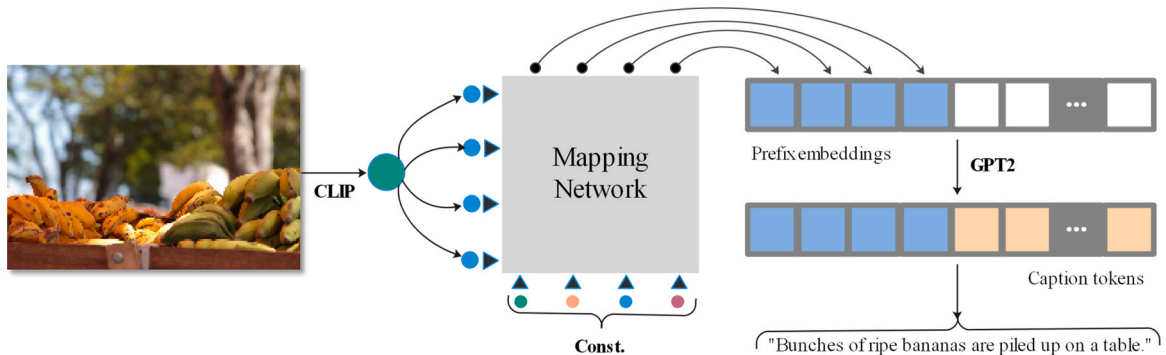


Fig. 1. Framework of the ClipCap.

language-image pre-trained model)[10], and ViECap (transferable decoding with visual entities for zero-shot image captioning) [11] fine-tune the image encoder, directly using its output as soft prompts for the LLM. Effectively utilizing frozen image encoders and LLMs for various vision-language tasks can achieve stronger performance at lower computational costs. Although mapping network-based methods require training only the mapping network, reducing training costs and enhancing generalization performance, they still face challenges in domain adaptation and limited generalization in specific tasks.

3. The proposed method

3.1. Overall framework

The overall framework of the proposed ClipCap++ is shown in Fig. 2. This is a model that integrates the CLIP visual encoder, few-shot key-value pair module, residual connection module, mapping network, and GPT-2 for image captioning. Based on the ClipCap model, the proposed model adds a key-value pair module and a residual connection module, named ClipCap++. The training process of ClipCap++ is as follows: First, the frozen CLIP model is used for visual information encoding. Due to the association between visual and text embeddings in CLIP, this association significantly reduces training resources and time costs. Then, information from the few-shot dataset stored in the key-value pair model is introduced, and its visual and label information are used to optimize feature. Next, pre-trained visual information is connected with optimized features to obtain joint embeddings, ‘clip_emb’, preventing the model from overfitting to few-shot data. Finally, the full-connected (FC) mapping network processes clip_emb to produce prefix_emb, a fixed-size sequence. This embedding, combined with title text embeddings, is input into the language model, which outputs textual description information. During inference, the language model starts from the prefix_emb and generates the title word by word. This solution bridges the gap between the visual and textual worlds by fine-tuning a simple mapping network. The mapping network acts as a bridge between the image and text spaces, responsible for mapping image

vectors to the text space, resulting in a prompt vector sequence. The mapping network consists of fully connected layers, ‘constant_embs’, and a transformer module. To ensure feasibility, we use GPT-2, which can generate corresponding word sequences through the above mapping process.

3.2. Training objective

The main architecture of the model in this paper is an encoder-decoder structure based on a mapping network. For a given series of image-text pairs in the training data $\{x^i, c^i\}_{i=1}^N$, where x^i represent the i -th image and c^i represent the text description of the i -th image. The training objective of the model is to generate a meaningful text description for a new input image. First, for the text description in each training sample, we pad the number of tokens to the maximum length $c^i = c_1^i, \dots, c_l^i$. The training objective is defined as follows:

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(c_1^i, \dots, c_l^i | x^i) \quad (1)$$

where θ represents the trainable parameters of the model.

Both the adversarial training of CLIP and the autoregressive pre-training of GPT-2 adopt similar concepts, learning representations by maximizing similarity measures. The rich semantic embeddings of CLIP, which contain the essential visual data, can serve as conditions for the model. In this paper, these conditions are considered as the prefix for generating text. Since the required semantic information is encapsulated in the prefix, we use the autoregressive language model GPT-2 to predict the next token. The training objective can be further described as follows:

$$\max_{\theta} \sum_{i=1}^N \sum_{j=1}^l \log p_{\theta}(c_j^i | x^i, c_1^i, \dots, c_{j-1}^i) \quad (2)$$

To extract visual information from image x^i , we first utilize the visual encoder of a pre-trained CLIP model to obtain visual encodings f_{test} .

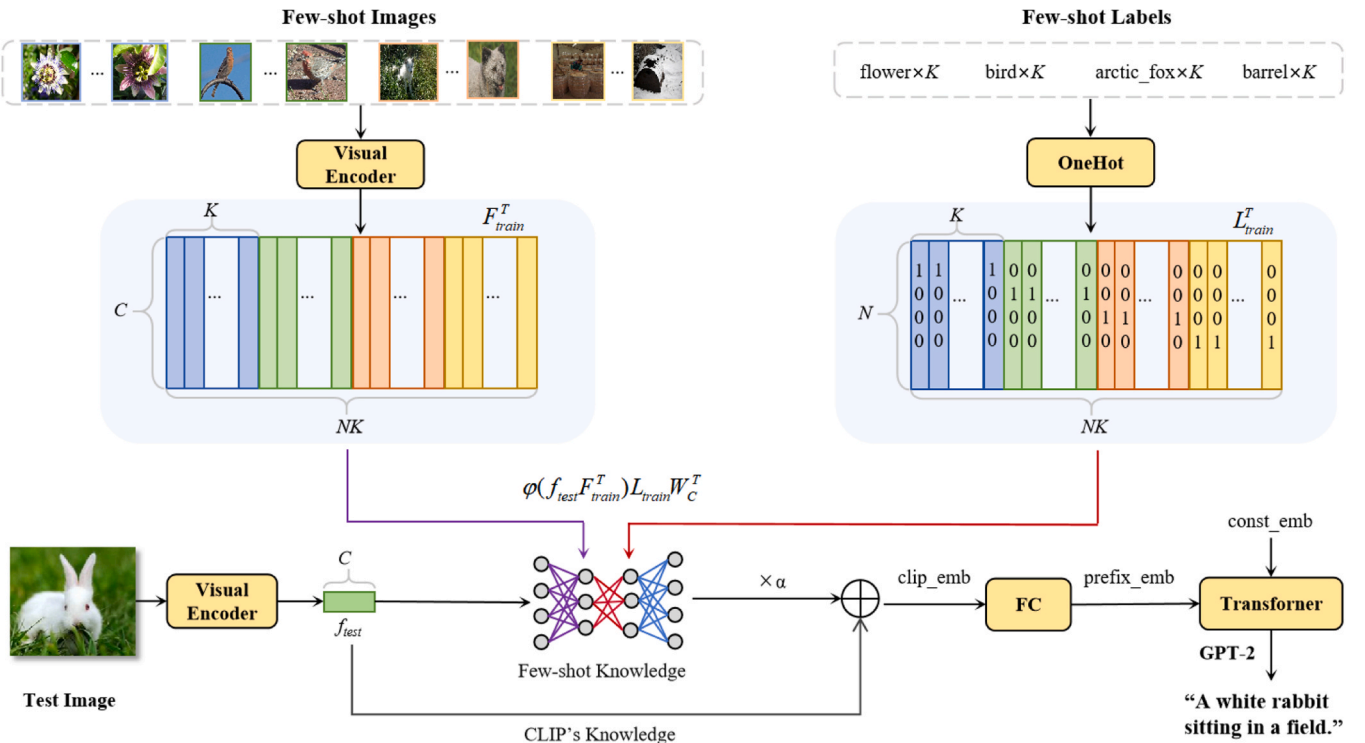


Fig. 2. Framework of ClipCap++.

Subsequently, these encodings undergo integration with few-shot data specific to a particular domain through key-value pair modules and residual connection modules, resulting in image vectors (clip_emb). Then, a full-connected (FC) mapping network is employed to generate k visual embedding vectors, each of p_j^i shares the same dimensionality as word embeddings. The processing pipeline described above is outlined as follows:

$$p_1^i, \dots, p_k^i = \text{FC}(\text{clip_emb}(x^i)) \quad (3)$$

Next, the obtained visual embeddings are concatenated with caption embeddings to obtain prefix vectors:

$$Z^i = p_1^i, \dots, p_k^i, c_1^i, \dots, c_l^i \quad (4)$$

During the training process, we provide the prefix vector $\{Z^i\}_{i=1}^N$ to the language model and use instruction fine-tuning to guide the language model to predict text tokens in an autoregressive manner, conditioned on the prefix. The designed instruction fine-tuning template is shown in Table 1:

Here, we utilize a simple and effective cross-entropy loss function to train the mapping network:

$$\mathcal{L}_x = - \sum_{i=1}^N \sum_{j=1}^l \log p_\theta(c_j^i | p_1^i, \dots, p_k^i, c_1^i, \dots, c_{j-1}^i) \quad (5)$$

3.3. Key-value module

Building upon the ClipCap model, this paper introduces a key-value pair module to extract necessary new knowledge from a few-shot dataset, thus updating the prior knowledge stored in CLIP. In the key-value pair module, the visual representation obtained by encoding images through CLIP is regarded as keys, while the label vectors obtained through one-hot encoding are regarded as values for these keys. The construction process of the key-value pair module is as follows:

$$\begin{aligned} F_{\text{train}} &= \text{VisualEncoder}(I_K) \\ L_{\text{train}} &= \text{OneHot}(L_N) \end{aligned} \quad (6)$$

where $F_{\text{train}} \in R^{N \times C}$ represents the visual representation obtained through pre-trained CLIP model, and $L_{\text{train}} \in R^{N \times K \times N}$ represents the label vectors obtained through one-hot encoding. In the given few-shot dataset, there are N classes of training samples for image classification, each class containing K annotated images I_K , and the class labels for images are represented as L_N .

The traditional approach of generating image captions using pre-trained Visual Language Models (VLM) and LLM often suffers from the issue of generating low-quality captions in specific domains. In our model, the CLIP module utilizes visually informed language to guide GPT-2 in generating captions related to images, while the key-value pair module enhances the model's ability to generate captions for specific domains by storing information from a small sample dataset. During inference, features generated from test examples are treated as queries, retrieving and aggregating information from the key-value pair module to optimize embedding representations. This is akin to the attention mechanism in Transformers, effectively improving the model's

capability to recognize specific information in specific domains for the image captioning task.

3.4. Residual connection module

To adjust the influence ratio of the key-value pair module on the overall model's visually informed language guidance, this paper introduces a residual connection module. Specifically, after the key-value pair module, a residual ratio is employed to adjust the weighting between the CLIP visual encoding and the optimized visual encoding. The adaptive adjustment of CLIP visual encoding can be achieved through matrix multiplications:

$$FL_emb = \varphi(f_{\text{test}} F_{\text{train}}^T) L_{\text{train}} W_C^T \quad (7)$$

where $f_{\text{test}} \in R^{1 \times C}$ represents the normalized features of the test image generated by the CLIP encoder, which undergo similarity calculation with key vectors in the key-value pair cache module; W_C is the transformation vector generated from the pre-trained text encoder; and $FL_emb \in R^{1 \times C}$ denotes the intermediate vector generated. The formula for calculating the correlation measure is as follows:

$$A = \exp(-\beta(1 - f_{\text{test}} F_{\text{train}}^T)) \quad (8)$$

where β is used to control the sharpness of the affinity; when β is large, training samples in the embedding space that are most similar to the test image have a greater impact on predictions. The exponential function converts similarity to non-negative values, yielding $A \in R^{1 \times NK}$.

Next, the introduction of the residual ratio is to adaptively adjust the weighting between CLIP visual encoding and the optimized visual encoding:

$$\text{clip_emb} = \alpha A L_{\text{train}} W_C + f_{\text{test}} \quad (9)$$

where α denotes the residual ratio, indicating the proportion attributed to the few-shot sample dataset. As α increases, the influence of the few-shot dataset becomes more pronounced. It is empirically suggested that in cases of substantial domain disparity between pre-trained and downstream few-shot tasks, a higher α is advisable, as it necessitates leveraging more knowledge from the few-shot sample dataset. Conversely, in scenarios with minimal domain discrepancy, a lower α suffices. Hence, experimental validation of appropriate values for α and β across diverse few-sample datasets becomes imperative for optimal model performance.

During the inference phase, when presented with a test image, we leverage the image embeddings derived from the CLIP visual encoder as hard prompts, while concurrently crafting nuanced soft prompts through the key-value pair module, enhancing the model's sensitivity to entities. Benefiting from the robust transferability ingrained in hard prompts sourced from unseen training data, our framework exhibits resilience to fluctuations in image domains. Moreover, by integrating entity-aware soft prompts, our model demonstrates heightened discernment in recognizing specific domains within images. An effective image captioning model should strike a balance between soft and hard prompts. Weak visual guidance often leads to modality shift issues, where language priors dominate the decoding process within LLM, resulting in descriptions unrelated to the corresponding images. Conversely, when the key-value pair cache module dominates, learning soft visual prompts can lead to overfitting, resulting in poor model performance in describing various objects (visual entities), potentially causing hallucinations in generated captions. The residual connection module proposed in this paper, through adjusting appropriate residual ratios (α) and sharpness (β), significantly mitigates issues such as modality shift and object hallucination when describing images containing new objects.

Table 1

Template for prompt.

Prefix	p_j^i (prefix_emb)
Task	Generate a descriptive text based on the given Prefix.
Description	The generated text should align with the visual information provided by the prefix, ensuring coherence and relevance.
Instruction	Use the Prefix as the starting point for the generated text. Ensure the text is relevant and informative based on the prefix context. Follow an auto-regressive approach to predict the next token in the sequence.
Output	[Text description]

4. Experiments

4.1. Datasets

This study performs comprehensive experiments to assess the caption generation performance of our proposed ClipCap++ model across three distinct benchmark datasets: COCO [22], NoCaps [23], and Conceptual Captions [24]. The detailed statistical characteristics of these datasets are presented in Table 2.

The COCO dataset containing 80 common object categories with moderate data volume, this dataset serves as our primary training set owing to its balanced representation. The NoCaps dataset contains only validation and test sets specifically designed to assess model generalization to novel classes, with model training conducted on the COCO dataset. Evaluations are categorized into three domains:

- In-domain: Contains images from classes present in the training set
- Near-domain: Comprises images of novel classes that exhibit visual similarity to known classes (e.g., 'wolves' vs. trained 'dogs')
- Cross-domain: Consists exclusively of images from completely novel classes without visual similarity to training classes (e.g., 'airplanes')

The Conceptual Captions dataset features greater stylistic diversity across multiple categories, providing a robust testbed for evaluating model generalization capabilities.

4.2. Evaluation metrics

To ensure a fair comparison with the state-of-the-art image captioning model ClipCap [1], we use BLEU [25], METEOR [26], CIDEr [27], SPICE [28], and ROUGE-L [29] as evaluation metrics for image caption generation.

- BLEU (BiLingual Evaluation Understudy): Measures the n-gram overlap between generated captions and reference captions, primarily used to assess grammatical accuracy.
- METEOR (Metric for Evaluation of Translation with Explicit Ordering): Considers syntax, semantics, and lexical matching comprehensively, providing a more holistic assessment of the quality of generated captions.
- CIDEr (Consensus-based Image Description Evaluation): A consensus-based metric that emphasizes the semantic consistency between generated and reference captions.
- SPICE (Semantic Propositional Image Caption Evaluation): Evaluates the semantic richness and accuracy of generated captions through semantic proposition analysis.
- ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation - Longest common subsequence): Measures the longest common subsequence between generated and reference captions, mainly used to assess semantic similarity.

The choice of evaluation metrics should be adjusted according to the characteristics of the dataset and the task objectives. For example, for the NoCaps dataset, metrics such as ROUGE-L should be used to assess the model's generalization ability to new classes and concepts. Specifically, ROUGE-L, CIDEr, and SPICE metrics are employed for the COCO dataset; CIDEr and SPICE metrics are used for the NoCaps dataset; and

BLEU, METEOR, ROUGE-L, CIDEr, and SPICE metrics are utilized for the Conceptual Captions dataset. We also record the training time and the number of hyperparameters for each method, as these factors are crucial for image caption generation. Optimizing training time allows for the rapid development of new models on new data, facilitates the creation of model ensembles, and reduces energy consumption. The number of trainable parameters is a common measure of a model's feasibility. Consistent with other studies, we report training time in GPU hours and specify the type of GPU used.

4.3. Experimental setup

In our model, we use ResNet-50 [30] as the visual encoder for the CLIP backbone and Transformer [31] as the text encoder. ResNet-50 is a relatively lightweight Convolutional Neural Network (CNN), suitable for rapid deployment on ordinary GPUs or CPUs. The key-value pair module uses the ImageNet dataset as the source of few-shot information. For prompt design, we adopt the prompt ensembling strategy from Tip-adaptor [32], inputting seven templates into the CLIP text encoder and averaging the results to obtain the final prompt. The prompt integration strategy enhances the model's generation capability by combining multiple prompt templates. The seven prompt templates are: "itap of a [CLASS]", "a bad photo of the [CLASS]", "a origami [CLASS]", "a photo of the large [CLASS]", "a [CLASS] in a video game", "art of the [CLASS]", and "a photo of the small [CLASS]". These prompt templates are partly derived from the original ClipCap model. We employ CLIP-style preprocessing by resizing the shorter side of the cropped images to 224 while maintaining the original aspect ratio.

The model hyperparameters are set as follows: learning rate = 0.001, augment epoch = 10, and training epoch = 20. The data augmentation techniques used in the training phase include random cropping, horizontal flipping, color jittering, and rotation. After conducting ablation experiments, we determined that the optimal values for α and β are 1.55 and 0.65, respectively. Considering experimental feasibility, the language model used is GPT-2, which includes an 8-layer Transformer network with 8 attention heads and a hidden vector size of 768 dimensions. The length of the learnable soft prompts is set to 10. During training, we freeze the CLIP text encoder. For the caption generation task, we use beam search with a beam size of 5. The settings of other parameters are consistent with those in the original ClipCap model.

We compare the proposed method ClipCap++ with the following image captioning approaches:

- Classical supervised methods, including BUTD [33], Oscar [34], and VLP [35]. BUTD uses LSTM for caption generation, while VLP and Oscar utilize BERT.
- State-of-the-art methods, including ViECap [36], I-Tuning [37] and ClipCap, which leverage GPT-2 for caption generation and CLIP for encoding.

As mentioned earlier, experiments are conducted on three widely used image captioning datasets: NoCaps, COCO, and Conceptual Captions. For the COCO dataset, we follow the commonly used Karpathy split [38] method to divide the dataset into training and test sets for performance evaluation. For the NoCaps dataset, in order to validate the model's transfer capability, we can train the model on one type of data set and test it on another type. For the Conceptual Captions dataset, we

Table 2
Statistics of the datasets.

	COCO [22]	NoCaps [23]	Conceptual Captions [24]
Data source	Flickr images	OpenImages images + novel objects	Web auto-crawled
# of images	330,000	15,100 (4500 validation images, 10,600 test images)	3300,000 (image URL, caption) pairs
# of categories	80 common categories	open domain	open domain
# of captions	5 human-written descriptions per image	5 human-written descriptions per image	1 auto-generated description per image

use all of its data as the test set for performance evaluation. All comparative experiments are conducted under the same hardware environment: GPU is NVIDIA GeForce RTX 1080 with 8 GB of memory, and CPU is AMD R7 7700X with 32 GB of memory and 1TB of SSD storage.

4.4. Experimental results

4.4.1. Comparative experiments

Tables 3 to 5 provide the quantitative evaluation results of our model, ClipCap+, compared to the baseline models on three datasets. Table 3 showcases the performance of different models in generating captions for in-domain images, Table 4 illustrates the performance for cross-domain images, and Table 5 presents the performance of three types of image captioning tasks on the NoCaps dataset. In each test category, the optimal performance metrics are highlighted in bold, while suboptimal results are marked with underlines. The row/column labeled "Improvement" indicates the performance improvement magnitude of our proposed ClipCap+ model compared to the baseline ClipCap model.

It can be seen from the table that in terms of accuracy, traditional methods perform well in in-domain image captioning tasks on the COCO dataset, but perform poorly on the NoCaps dataset, indicating that the performance of traditional methods is unstable. Overall, the performance of our proposed model ClipCap+ exceeds that of traditional image captioning methods (BUTD, OSCA, and VLP) and the current state-of-the-art methods (ViECap, I-Tuning, ClipCap). This indicates that the two major improvement measures proposed in this paper, namely the key-value pair module and the residual connection module, have a positive effect on image captioning generation. In terms of training time and parameter size, our model ClipCap+ achieves comparable results to the state-of-the-art image captioning models, with the training time reduced by several orders of magnitude compared to traditional image captioning models. This is even the case when using a lower-level GPU, which demonstrates that our model is an efficient image captioning model. In terms of transferability, ClipCap+ exhibits superior performance in near-domain and out-of-domain tasks. This indicates that the proposed model has excellent transferability.

Additionally, we performed paired t-tests on the NoCaps dataset to evaluate performance differences between our proposed ClipCap+ model and the baseline ClipCap model across multiple experimental categories and evaluation metrics. Under identical testing conditions, both models were executed 10 times, with all metric scores recorded for subsequent calculation of means and standard deviations prior to conducting two-tailed t-tests. As presented in Table 6, ClipCap+ demonstrates statistically significant superiority over the baseline model ($p < 0.05$) across all experimental categories and evaluation metrics, with performance improvements ranging from 0.34 to 1.85 points. These consistent and statistically significant enhancements confirm the effectiveness of our proposed model.

4.4.2. Ablation experiment

In this work, we introduce two novel optimizers - the Key-Value (KV) module and the Residual (Res) module- built upon the baseline ClipCap classifier. To evaluate the impact of the KV and the Res in image

Table 4

Performance comparison for cross-domain image captioning on the Conceptual Captions dataset.

Model	ROUGE-L	CIDEr	SPICE	Time
VLP	24.35	<u>77.57</u>	<u>16.59</u>	1200 h(V100)
ClipCap	<u>25.12</u>	71.82	16.07	72 h(GTX1080)
ClipCap+ +	27.10	80.21	16.90	72 h(GTX1080)
Improvement	7.88 %	11.68 %	5.16 %	

captioning, we conducted ablation experiment on the COCO dataset with the following configurations:

- (1) ClipCap (baseline): The original model without any modifications.
- (2) ClipCap+KV: Incorporates the KV module while excluding the Res module, maintaining a fixed 1:1 ratio between KV and CLIP.
- (3) ClipCap+KV+Res (our model): Integrates both modules.

It should be noted that the "ClipCap+Res" configuration was not included in the ablation study because the Res module is specifically designed to mediate the fusion between the KV module's output and CLIP's prior information, and its functionality fundamentally depends on the presence of the KV module. Therefore, the Res module cannot be directly integrated with the baseline model. The experimental results are shown in Table 7. By comparing the experimental results of ClipCap and ClipCap + KV, we found that the key-value pair module significantly enhanced the category relevance of the generated text by incorporating domain-specific knowledge. The comparison between ClipCap + KV and ClipCap + KV + Res shows that the residual connection module further optimized the knowledge fusion process by dynamically adjusting the weights between the KV module and the CLIP prior information.

4.4.3. Parameter sensitivity analysis

To validate the impact of model parameters on image captioning performance, we conducted a series of experiments. Fig. 3 illustrates the effect of varying the residual ratio (α) and affinity sharpness (β) on the accuracy of image captioning. From the figure, it can be observed that when β is less than 0.31, increasing the value of α while keeping β constant results in a gradual rise in accuracy. The accuracy ceases to improve once α reaches 1.55. Subsequently, maintaining α constant and increasing β leads to a further improvement in accuracy, which stabilizes when β approaches 0.31. Further increasing α beyond this point does not affect the model's accuracy, indicating that the model has reached its optimal stable state. Therefore, the optimal values for α and β were determined to be 1.6 and 0.31, respectively, for the experiments conducted in this study.

Table 8 illustrates the impact of the number of image categories used in few-shot fine-tuning on the accuracy of image captioning. From the table, it can be observed that initially, as the number of image categories used in fine-tuning increases, the performance of the model gradually improves. However, when the number of categories increases to 16, further increasing the number of categories does not lead to a performance improvement, indicating that the model training has reached saturation at this point. Therefore, in the experiments conducted in this study, the number of image samples used in the fine-tuning stage was set to 16.

Table 9 demonstrates the influence of the learning rate (lr) on the accuracy of image captioning. From the table, it can be observed that when lr is set to 0.1, the parameter update step size is too large, leading to the model skipping over the local minimum of the loss function during training, resulting in abnormal accuracy values. Conversely, when lr is set to 0.0001, the model convergence speed is too slow, and within a limited time frame, the optimal model performance cannot be achieved. However, when lr is set to 0.001, the model achieves the best performance. Therefore, in the experiments conducted in this study, the

Table 3

Performance comparison for in-domain image captioning on the COCO dataset.

Model	B@4	METEOR	CIDEr	SPICE	Time
BUTD	36.20	27.00	113.50	20.30	960 h(M40)
VLP	36.50	28.40	<u>117.70</u>	21.30	48 h(V100)
Oscar	36.58	30.40	124.12	23.17	74 h(V100)
I-Tuning	34.80	28.30	116.70	21.80	6 h(GTX1080)
ClipCap	33.53	27.45	113.08	21.05	6 h(GTX1080)
ClipCap+ +	35.10	<u>29.00</u>	114.20	<u>22.04</u>	6 h(GTX1080)
Improvement	4.68 %	5.65 %	0.99 %	4.70 %	

Table 5

Performance comparison for cross-domain image captioning on the NoCaps dataset.

Category	Metric	BUTD	Oscar	ViECap	I-Tuning	ClipCap	ClipCap+ +	Improvment
in-domain	CIDEr	74.30	79.60	61.10	83.90	<u>84.85</u>	86.70	2.18 %
	SPICE	11.50	12.30	10.40	<u>12.40</u>	12.14	13.12	8.07 %
near-domain	CIDEr	56.90	66.10	64.3	70.30	66.82	<u>67.81</u>	1.48 %
	SPICE	10.30	<u>11.50</u>	9.90	11.70	10.92	<u>11.26</u>	3.11 %
cross-domain	CIDEr	30.10	45.30	65.00	48.10	<u>49.14</u>	49.88	1.51 %
	SPICE	8.10	<u>9.70</u>	8.60	9.50	9.57	10.01	4.60 %
overall	CIDEr	54.30	63.80	66.20	<u>67.10</u>	65.83	67.31	2.25 %
	SPICE	10.10	11.20	9.50	<u>11.40</u>	10.86	11.80	8.66 %
	Time	960 h	74 h	6 h	6 h	6 h	6 h	

Table 6

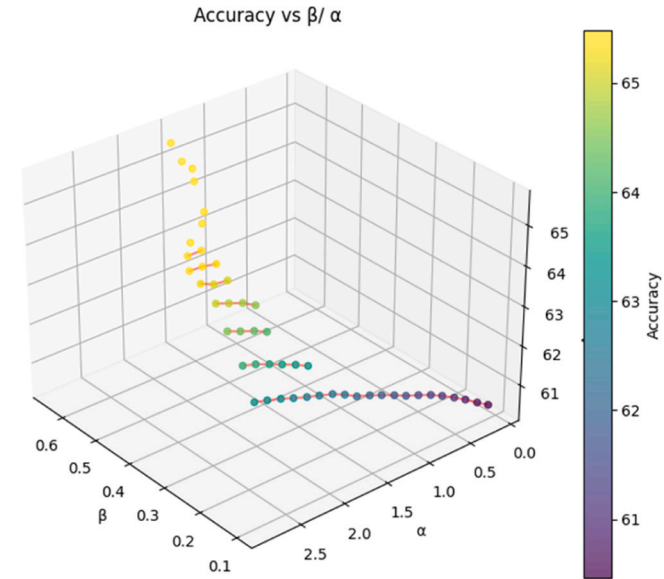
Results of significance testing.

Category	Metric	ClipCap Mean	ClipCap+ + Mean	Mean Difference	95 % confidence interval	p-value
in-domain	CIDEr	84.85 ± 0.75	86.70 ± 0.52	1.85	[1.12, 2.58]	0.0004
	SPICE	12.14 ± 0.51	13.12 ± 0.34	0.98	[0.62, 1.34]	0.0008
near-domain	CIDEr	66.82 ± 0.57	67.81 ± 0.67	0.99	[0.53, 1.45]	0.0012
	SPICE	10.92 ± 0.35	11.26 ± 0.65	0.34	[0.08, 0.60]	0.0120
cross-domain	CIDEr	49.14 ± 0.11	49.88 ± 0.87	0.74	[0.31, 1.17]	0.0020
	SPICE	9.57 ± 0.23	10.01 ± 0.92	0.44	[0.13, 0.75]	0.0060
overall	CIDEr	65.83 ± 0.54	67.31 ± 0.52	1.48	[0.83, 2.13]	0.0009
	SPICE	10.86 ± 1.15	11.80 ± 0.87	0.94	[0.51, 1.37]	0.0010

Table 7

Results of the ablation experiments on the COCO dataset.

Model	BLEU-4	METEOR	CIDEr	SPICE
ClipCap	33.53	27.45	113.08	21.05
ClipCap+KV	34.12	28.23	113.65	21.34
ClipCap+KV+Res	35.10	29.00	114.20	22.04

**Fig. 3.** The effect of residual ratio (α) and affinity sharpness (β) on image captioning accuracy.**Table 8**

Impact of the number of image categories.

Shots	1	2	4	8	16	32
Accuracy	61.16	61.76	62.54	63.94	65.51	65.46

Table 9

Impact of the learning rate (lr).

lr	0.01	0.001	0.0001	0.002
Accuracy	65.04	65.51	64.14	65.26

learning rate for model fine-tuning training was uniformly set to 0.001.

Fig. 4 illustrates the impact of the length of the input prefix given to GPT-2 on the performance of image captioning. In the experiment, we evaluated the model based on two performance metrics, BLEU@4 and CIDEr, on both the training and testing datasets. BLEU is utilized to assess the consistency between the generated text and the images, while CIDEr measures the accuracy of image descriptions. The results indicate that when the prefix length is too short, the model performance deteriorates due to insufficient information to accurately describe the image. Conversely, an excessively long prefix length leads to overfitting, resulting in decreased performance. Through experimentation, we observed that a prefix length of 10 yields the best performance. Consequently, we choose to set the length of the given prefix to 10.

4.4.4. Visualization analysis

Table 10 presents the image captioning results of our model ClipCap+ +, the baseline model ClipCap, and the multimodal large language model GPT-4 and CogVLM2-Llama3 [39] on specific categories (tissues) within the ImageNet dataset. Incorrectly generated objects are highlighted in red, while correctly generated objects are highlighted in blue. From the table, it is evident that the ClipCap model tends to generate incorrect information, possibly due to its training process not involving images of the tissues category. Although GPT-4 and CogVLM2-Llama3 can generate more detailed textual descriptions, it also fails to recognize the specific entities within this domain because it is a pretrained model that has not been fine-tuned for this specific domain. Our model, ClipCap+ +, achieves high accuracy in generating captions within this domain, thanks to the introduction of the key-value pair module. This module leverages few-shot information to fine-tune the model, making it familiar with new domain knowledge and enabling it to generate more accurate textual descriptions for images in new domains.

In the practical deployment of ClipCap+ +, the GPT-2 generation component emerges as the primary bottleneck. In fields where precision

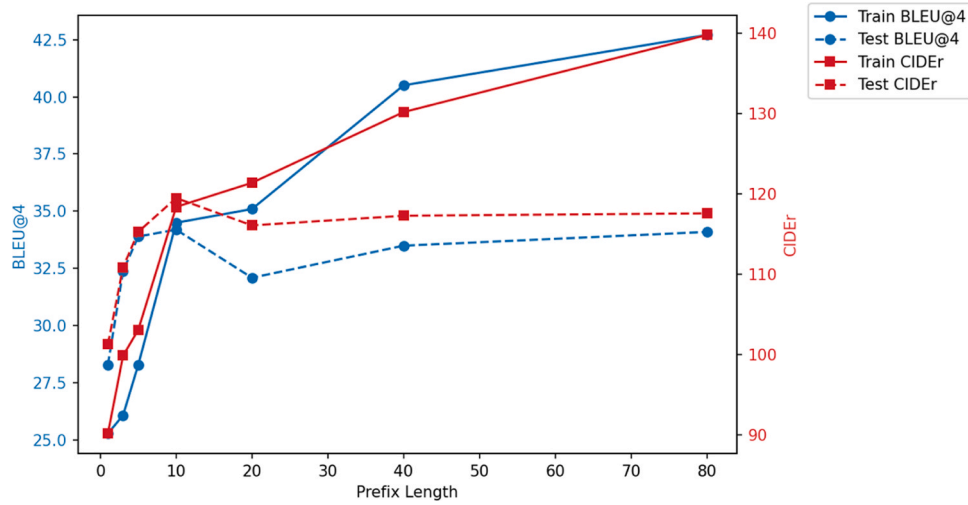


Fig. 4. The effect of prefix length on image captioning performance.

Table 10
Image captioning results on the ImageNet dataset.

	ClipCap	CogVLM2-Llama3	GPT-4	ClipCap++
	A smiling woman smiling while holding a wedding cake.	A moment of joy and surprise on the bride's wedding day.	A beaming bride in a white gown and veil laughs, capturing a moment of joy.	A smiling woman holding a tissue with a smile.
	A person standing in front of a toilet.	A person is on the toilet, feet on the seat, likely pulling down toilet paper from the wall-mounted roll.	A ghostly double exposure captures feet standing on a toilet in a stark, illuminated room.	A man stands on the toilet taking out tissues.
	A bunch of different types of packaged beverages in a store.	A shelf lined with various household cleaning products.	A neatly organized shelf displays 'Skip' detergent boxes with clear price tags in a simple, retail setting.	A pile of different types of tissues are in a store.
	A man riding a bicycle down a dirt road.	A man is pushing a bicycle on a dirt road surrounded by lush greenery.	A person rests on a bicycle, poised on a dusty road flanked by greenery, under hazy sunlight.	A man rides down a dirt road on his bicycle with tissues hanging on it.

is not a critical requirement, it is advisable to employ quantization techniques or opt for lightweight model alternatives. The system compatibility is good in the cloud, but edge devices require model compression, such as trimming the KV sample library. In industrial scenarios such as warehouse inventory management, the explainability of the system's decision-making relies on the sample retrieval mechanism of the KV module, particularly the matching and tracking of historical records of goods entering and leaving the warehouse. It is necessary to provide similarity visualization and dynamic adjustment

interfaces for the residual ratio (α/β) to balance the weight of domain knowledge and reduce the risk of hallucination. When real-time requirements are high, it is recommended to combine hardware acceleration with a manual review process.

5. Conclusion

With the continuous development in the field of artificial intelligence, integrating pre-trained models such as CLIP into tasks like image captioning using the concept of transfer learning is crucial. However, existing mapping network-based models like ClipCap face challenges in adapting to specific domains. To address this issue, we propose the ClipCap++ model, which incorporates key-value pair and residual connection modules. These enhancements endow the model with transfer ability, effectively improving its accuracy in generating image captions in specific domains. Currently, more powerful language models such as GPT-4 and DeepSeek have emerged and equipped with multimodal capabilities to perform a wide range of tasks. For future work, we will explore the utilization of more advanced and better-performing language models for image-text generation. A key challenge in integrating models like GPT-4 is aligning visual and textual representations to mitigate semantic gaps (modality shift). GPT-4's multimodal architecture allows it to process images and text simultaneously, which can help mitigate this challenge. However, ensuring seamless integration requires careful alignment of the visual and textual embeddings. We plan to explore techniques such as prompt engineering and fine-tuning to optimize the model's performance in multimodal tasks.

CRedit authorship contribution statement

Ye Wu: Writing – original draft. **Zhenzhen Sheng:** Writing – review & editing, Formal analysis. **Ruiqin Wang:** Validation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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